
INTRODUCTION TO ENGINEERING · LAB SESSION 1 (Week 1)

Engineering as a Profession

The Practical Lab



Traits, ethics, and the boundary between science, engineering, and design — put to the test in three hands-on activities.

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Teaching Assistant · Introduction to Engineering · Fall 2026

Builds directly on Lecture 1: “Engineering as a Profession,” Dr. Mohammad Reza Bahrami — 80-minute lab, 12-15 students.

About Me

- **Assistant Professor**

Introduction to Engineering — Tashkent University of Technology

- **MSc / PhD track**

Mechanical & Management Engineering, Politecnico di Milano

- **Research interests**

Lean manufacturing, sustainability, and quantitative productivity assessment
Google Scholar: scholar.google.com (search “Ibrokhimjon Khamidov”)

- **Why I'm running this lab**

Ideas stick when you argue about them with your hands — not just read them



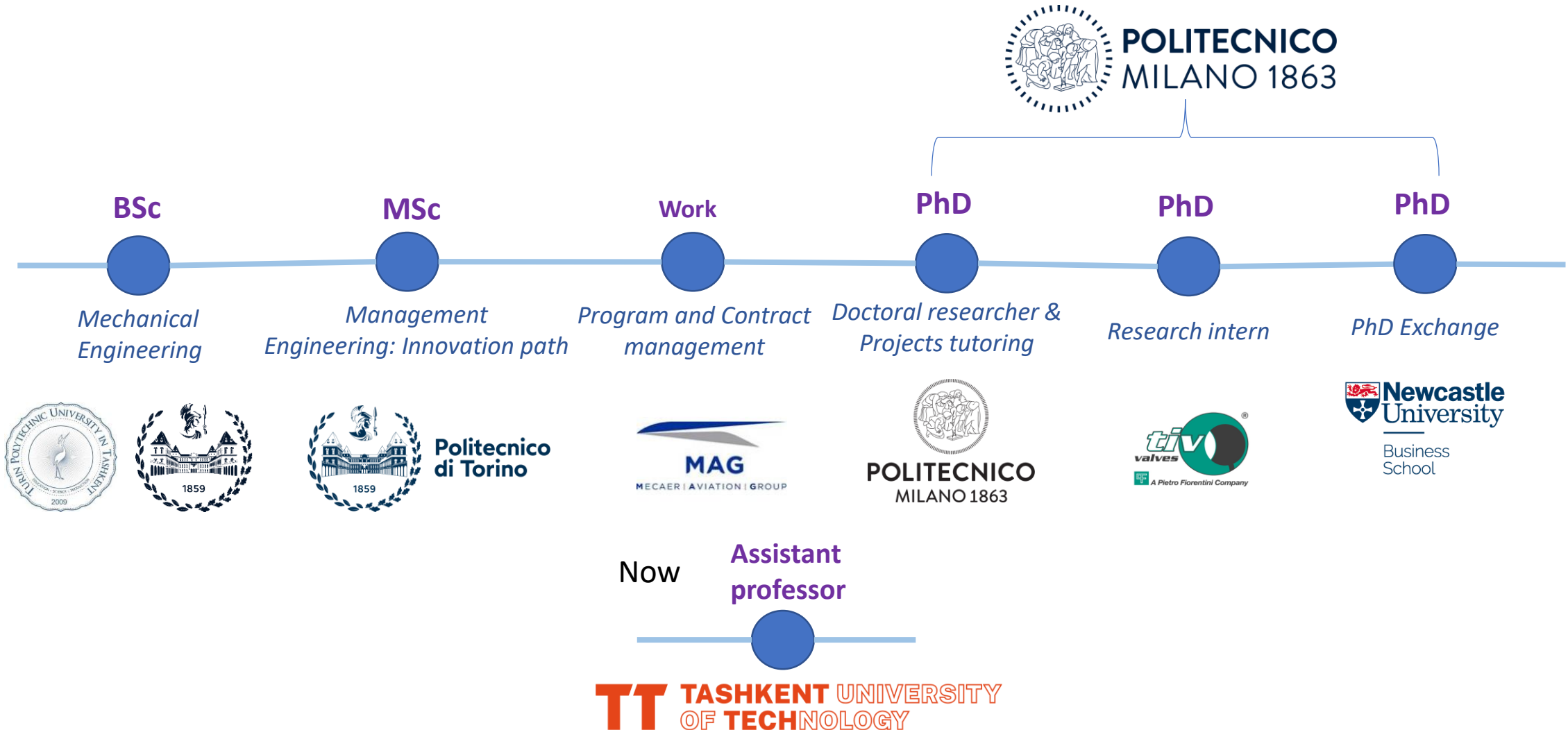
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My path to Engineering and Management



Lab Agenda — 80 Minutes



Recap: Lecture 1 Highlights

Key ideas + the 10 problem-solver traits

20 min



Form Your Groups

Choose teammates; we balance for trait diversity

10 min



Game 1 — The Ethics Courtroom

Argue real dilemmas against the NSPE code

25 min



Game 2 — Sort the Scenarios

Science, Engineering, Design, or Overlap?

25 min



Wrap-Up

What stuck, and where this goes next

PART 1



Recap — Lecture 1 Highlights

The vocabulary and mindset this whole lab runs on.

Quick Recap: What We Covered



Etymology

“Engineer” comes from Latin roots meaning cleverness and invention.



Historical role

Engineering has shaped every era, from ancient fortifications to the digital age.



Social role

Engineers sit at the interface between science and society — with real responsibility.



Economic role

One of the largest, best-paid professions worldwide — and a launchpad to other careers.



Culture & ethics

Shared traits, professional societies, and a code of ethics unite engineers globally.



Science vs. design

Science explains what is; design shapes possibilities; engineering builds what works.

Pick Your Trait

- Look back at the 10 traits — choose the one that feels most like you today.
- Write it on a sticky note or the back of your hand — low tech, high honesty.
- No wrong answers. This is about self-awareness, not judgment.



Why it matters

In a minute you'll form your own team. Knowing your own trait — and seeing your teammates' — is what lets us check the group isn't accidentally all “Creative” and no “Vigilant.”

The Mindset of a Good Problem Solver

The same 10 traits from Lecture 1 — in a minute, you'll pick the one that's really you.



Objective



Analytical



Creative



Dedicated



Curious



Courageous



Adventurous



Enthusiastic



Patient



Vigilant

Form Your Groups

- Self-select into groups of 4–5 — sit together now.
- Say your chosen trait out loud to your new group.
- I'll circulate and check each group's mix of traits.
- If a group is too one-note, I'll ask one person to swap groups — nothing personal, just balance.



Your task

A group of five “Objective”s will out-argue each other forever and never decide. A mix of traits is what makes the next two games actually work — that's the whole point of the swap.

PART 2 • 25 MINUTES



Game 1 — The Ethics Courtroom

Real dilemmas, the NSPE code, and a jury of your classmates.

The Ethics Courtroom

1

Get your cases

Each group receives 3–4 dilemma cards — short, real-feeling scenarios engineers actually face.

2

Deliberate (10 min)

Discuss each case as a group. Decide: what should this engineer do — and which canon of the code applies?

3

Present your verdict (10 min)

Each group presents one case in 2 minutes: the dilemma, your verdict, and the canon number you're citing.

4

The class is the jury

After each verdict, the room votes by show of hands: agree, disagree, or “it depends.” We discuss any real splits.

10 Canons of the Code (NSPE, adapted)

1

Safety first

Hold paramount the safety, health, and welfare of the public.

3

Speak truthfully

Issue public statements only in an objective, truthful manner.

5

No deception

Avoid deceptive acts of any kind.

7

No hidden payments

Don't accept pay from multiple interested parties without full consent.

9

Give proper credit

Acknowledge the real contributions of others.

2

Stay in your lane

Perform services only in areas of your competence.

4

No conflicts of interest

Act as a faithful agent; avoid personal interest competing with duty.

6

Act honorably

Conduct yourself responsibly, to protect the profession's reputation.

8

Don't misrepresent yourself

Never falsify your qualifications, experience, or results.

10

Don't stay silent

Report or refuse to approve code violations, even under pressure.

Let's Try One Together First

SAMPLE CARD

“A civil engineer is asked to sign off on a building's electrical wiring plan because ‘he's the only one free right now,’ even though he's never studied electrical systems.”

Ask the room: what should he do, and which canon applies?

[90 seconds — take 2–3 answers before revealing]

MODEL ANSWER

Decline, and refer the electrical work to a qualified electrical engineer.

Canon cited:

#2 — Stay in your lane: perform services only in areas of your competence.

This is the pattern every group will follow: verdict + canon number.

PART 3 • 25 MINUTES



Game 2 — Sort the Scenarios

Science, Engineering, Design, or all of the above?

Science, Engineering, Design — or Overlap?

1

Shuffle & deal

Each group gets a shuffled deck of scenario cards — one-sentence descriptions of real technical or creative work.

2

Sort into four piles (15 min)

Label four spaces on your table: Science, Engineering, Design, Overlap. Sort every card into one.

3

Compare answers (7 min)

We reveal the intended answers together. Every group keeps score — no penalty for a wrong sort, just discussion.

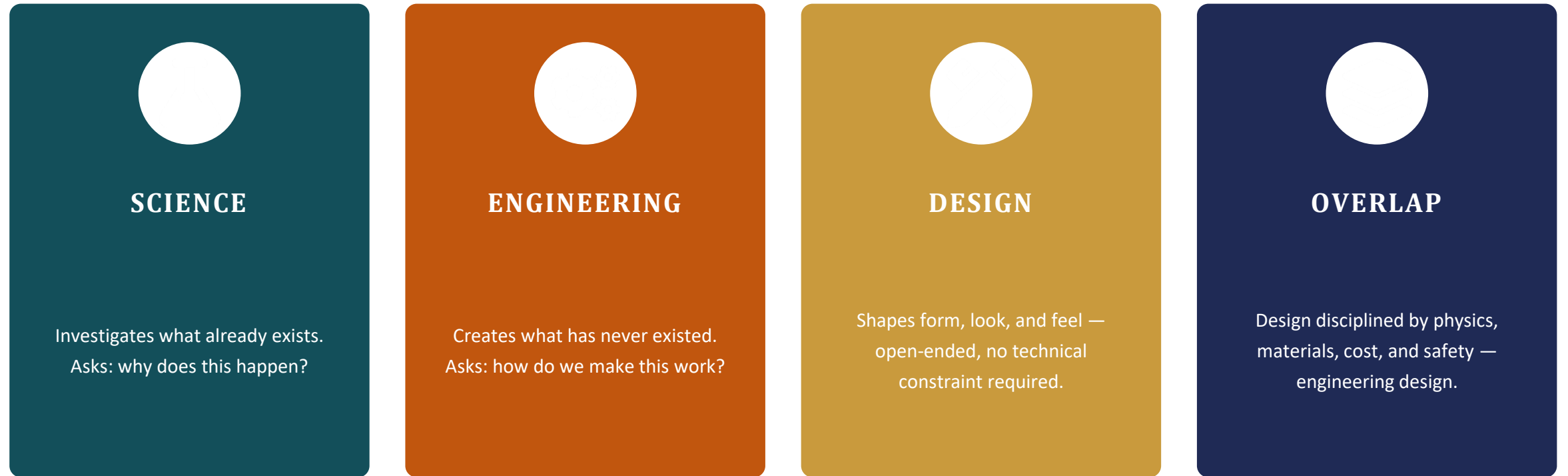
4

Defend the tricky ones

For any card your group got wrong, you get 30 seconds to argue why your answer was reasonable anyway.

KEEP THIS VISIBLE WHILE SORTING

The Four Buckets



Tie-breaker question: does the card involve shaping form or appearance? No → Engineering. Yes, no technical constraint → Design. Yes to both → Overlap.



Thank You — Great Work Today

What's one idea from today you'll actually remember next week?

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In case you have questions please write email
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